

Spring 2015

Problem 1 (Sp15). 1. Evaluate the integral $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$.

2. Prove that $0 < \frac{22}{7} - \pi < \frac{1}{256}$.

Problem 2 (Sp15). Suppose that g is a positive real valued function on the real numbers. If $a < b$ are real numbers, show that there is a finite sequence $a = t_0 < t_1 < \cdots < t_n = b$ of real numbers such that in each interval $[t_k, t_{k+1}]$ there is a point where the value of the function g is greater than the length of the interval.

Problem 3 (Sp15). 1. Describe all sets of reals that can be the image of the real line under a polynomial with real coefficients.

2. Find the image of the real plane under the polynomial $x^2 + (xy - 1)^2$.

3. Describe all sets of reals that can be the image of the real plane under a polynomial in two variables with real coefficients.

Problem 4 (Sp15, Fa20). Write two different Laurent series in powers of the complex variable z for the function

$$f(z) = \frac{1}{z(1+z^2)}$$

Give the domain of each of these series.

Problem 5 (Sp15). Compute the difference

$$\int_{|z|=3} \frac{e^{\pi/z} dz}{z^2 + 4} - \int_{|z|=1} \frac{e^{\pi/z} dz}{z^2 + 4}$$

where both integrals are taken in the counterclockwise direction.

Problem 6 (Sp15). Fix $n \geq 1$. Let $s = (s_1, \dots, s_n)$ and $t = (t_1, \dots, t_n)$ be $2n$ distinct complex numbers. Define the $n \times n$ matrices $C(t, s)$, $P(t, s)$ and $Q(s)$ with P and Q diagonal to have entries

$$C(t, s)_{ij} = \frac{1}{t_i - s_j} \quad P(t, s)_{ii} = \prod_{k=1}^n (t_i - s_k) \quad Q(s)_{jj} = \prod_{k \neq j} \frac{1}{s_j - s_k}$$

Show that $p(t) = P(t, s)C(t, s)Q(s)p(s)$, where p is any polynomial of degree less than n , and for a vector $r = (r_1, \dots, r_n)$, $p(r)$ is defined to be the vector $(p(r_1), \dots, p(r_n))$.

Problem 7 (Sp15). Compute the determinant

$$\Delta_n = \begin{vmatrix} \binom{0}{0} & \binom{1}{0} & \binom{2}{0} & \binom{n-1}{0} \\ \binom{1}{1} & \binom{2}{1} & \binom{3}{1} & \binom{n}{1} \\ \binom{2}{2} & \binom{3}{2} & \binom{4}{2} & \binom{n+1}{2} \\ \binom{n-1}{n-1} & \binom{n}{n-1} & \binom{n+1}{n-1} & \binom{2n-2}{n-1} \end{vmatrix}$$

Problem 8 (Sp15). Factor the polynomial

$$11x^5 - 11x^4 + 14x^2 - 21x + 7$$

into irreducible polynomials in $\mathbb{Q}[x]$.

Problem 9 (Sp15). Find a product of cyclic groups that is isomorphic to the group

$$(\mathbb{Z}_{12} \times \mathbb{Z}_{12}) / \langle (2, 6) \rangle$$

Problem 10 (Sp15). For all integers $n > 2$ prove the inequality

$$\frac{n^n}{e^{n-1}} < n! < \frac{(n+1)^{n+1}}{e^n}$$

Problem 11 (Fa94, Sp98, Sp15). Find the maximum area of all triangles that can be inscribed in an ellipse with semiaxes a and b , and describe the triangles that have maximum area.

Note: See also ??.

Problem 12 (Sp15). Suppose that

$$\begin{array}{ccccccc} a_{1,1} & + & a_{1,2} & + & \cdots & & \\ a_{2,1} & + & a_{2,2} & + & \cdots & & \\ & & \vdots & & & & \\ a_{m,1} & + & a_{m,2} & + & \cdots & & \\ & & \vdots & & & & \end{array}$$

is a countable collection of convergent series of nonnegative real numbers. Show that there is a convergent series $x_1 + x_2 + \cdots$ of real numbers converging slower than any of the given series, in the sense that for any m we have $x_n \geq a_{m,n}$ for all sufficiently large n .

Problem 13 (Sp95, Sp05, Sp15). Prove that there is no one-to-one conformal map of the punctured disc $G = \{z \in \mathbb{C} \mid 0 < |z| < 1\}$ onto the annulus $A = \{z \in \mathbb{C} \mid r < |z| < R\}$.

Problem 14 (Sp09, Sp15). Show that if $f : \mathbb{C} \rightarrow \mathbb{C} \cup \infty$ is a meromorphic function in the plane, such that there exist $R, C > 0$ so that for $|z| > R$, $|f(z)| \leq C|z|^n$, then f is a rational function.

Problem 15 (Sp15). What is the maximal dimension of subspaces in $\mathbb{R}^4$ on which the quadratic form $x_1x_2 - 3x_2^2 + x_3^2 + 2x_2x_4 + x_4^2$ is positive definite?

Problem 16 (Sp15, Fa16, Sp17). Let $\mathbb{D}$ be the unit disc in the complex plane $\mathbb{C}$, $f : \mathbb{D} \rightarrow \mathbb{C}$ an analytic function with

$$|f^{(k)}(0)| \leq M$$

for all $k \geq 0$, and let $t_i, s_i \in \mathbb{D}$ for $1 \leq i \leq n$. Define the matrix $A_{ij} = f(t_i s_j)$ for $1 \leq i, j \leq n$. For each $r \geq 1$ find an $n \times n$ matrix B with rank $\leq r$ and

$$|A_{ij} - B_{ij}| \leq \frac{2M}{r!}$$

for $1 \leq i, j \leq n$.

Problem 17 (Fa11, Sp15, Sp26). Let p be a prime number and G be a group such that $g^p = 1$ for all $g \in G$. Show that if $p = 2$ then G is abelian, and give an example with $p > 2$ where G is not abelian.

Problem 18 (Sp15). Let p be a prime, $p^{a(n)}$ the largest power of p dividing $n!$ and let $b(n)$ be the sum of the digits of n in base p .

1. Show that $a(n) = \lceil n/p \rceil + \lceil n/p^2 \rceil + \lceil n/p^3 \rceil + \cdots$ where $\lceil x \rceil$ is the largest integer at most equal to x .
2. Express $a(n)$ in terms of the digits d_k of the base p expansion $n = \sum d_k p^k$ of n (where $0 \leq d_k < p$).
3. Find a nontrivial linear relation between the functions n , $a(n)$ and $b(n)$, with coefficients that may depend on p but do not depend on n .